

Be the Judge

Judging explanations — not answers.

Name: _____ Date: _____

A right answer and a good explanation are not the same thing. Below, three people answer the same question. **All three get the right answer** — so you can't judge them on that. You have to judge something harder: how well each one explains *why*.

Give each answer a score from 1 to 3:

- 1 = Just the answer — no reason.
- 2 = Gives a reason — but only tries one way.
- 3 = A real explanation — it works no matter what you try.

Part 1 — Watch a judge at work

Maya has 7 socks. Can she split them into two equal piles — the same number in each?

Answer A

“No.”

Score it (circle one):

1

2

3

***Judge:** True — but it tells us nothing. How does she know? An answer with no reason isn't an explanation yet.*

Answer B

“No. I tried 3 in one pile and 4 in the other, and they weren't equal.”

Score it (circle one):

1

2

3

***Judge:** Better — it gives a real reason. But she only tried one way. Maybe a different split works? This shows one try failing; it doesn't show that every try must fail.*

Answer C

“No. If the two piles were equal, each would hold the same number — so the total would be that number doubled, and a doubled number is always even. But 7 is odd. So equal piles are impossible — there's no split to try.”

Score it (circle one):

1

2

3

***Judge:** This one wins. Notice it never tries a single split — it shows that two equal piles would always need an even total, and 7 is odd. That rules out every way at once. Read it, and no one could talk you out of it.*

Part 2 — Now you be the judge

Same scores: **1** = just the answer, **2** = a reason but only one try, **3** = a real explanation. Circle one for each.

Sam has 9 stickers. Can he give the same number to each of his 2 cousins, with none left over?

Answer A

“No. I gave 4 to one cousin and 5 to the other, and that’s not the same.”

Score it (circle one): **1** **2** **3**

Answer B

“No. If both cousins ended up with the same number, the total would be that number doubled — always an even number. But 9 is odd. So it can never split into two equal shares.”

Score it (circle one): **1** **2** **3**

Answer C

“No.”

Score it (circle one): **1** **2** **3**

Part 3 — You be the judge again

A brand-new question — and **four** answers this time. Score each one 1, 2, or 3 — the same score can go to more than one answer.

When you add two numbers — any two numbers — does the order ever change the total?

Answer A

“No. I’m completely sure, because I’ve added tons of numbers before and I thought about it very carefully, and swapping them around just doesn’t do anything that could ever make the total come out different.”

Score it (circle one): **1** **2** **3**

Answer B

“No. Adding means putting two groups together into one pile, and the pile is the same size whichever group you start with. So the order can never change the total, for any two numbers.”

Score it (circle one): **1** **2** **3**

Answer C

“No. I checked $6 + 9 = 15$ and $9 + 6 = 15$ — same answer both times.”

Score it (circle one): **1** **2** **3**

Answer D

“No.”

Score it (circle one): **1** **2** **3**

Answer Key — for parents

What this sheet teaches. The difference between getting the right answer and explaining it. Every answer here is correct — the whole exercise is judging the *explanations*, so the boys have to look at reasoning instead of results.

The scores. Part 1 is worked on the sheet ($A \rightarrow 1$, $B \rightarrow 2$, $C \rightarrow 3$). From Part 2 on the order is *shuffled* — if they score by length or position instead of reading, it will show immediately.

Part 2: $A \rightarrow 2$ (tries one split), $B \rightarrow 3$ (the real explanation: equal shares need an even total, 9 is odd), $C \rightarrow 1$ (no reason).

Part 3: $A \rightarrow 1$, $B \rightarrow 3$ (adding is combining two groups — same pile either way), $C \rightarrow 2$ (checks one example), $D \rightarrow 1$. Yes, A gets a 1 despite all those words: it never gives a reason, it just repeats the answer with confidence.

Two to dwell on. The one-example answers (Part 2 A, Part 3 C) are the mistake from the last test — showing that one particular attempt fails and thinking that settles it. A real explanation has to rule out *every* attempt, not just one. And Part 3 A is the other trap: lots of words, total confidence, zero reason. Sounding sure isn't explaining.

What you're listening for. When they say why the winner wins, you want some version of "it gives a reason that covers every case, not just one try." If they say that — in any words — they've got it.

How to run it. However you like — out loud together, or on paper. If writing is a slog, let them *say* the verdicts and you scribe; the judging is the point, not the handwriting.